AI-Augmented SOC Implementation for Enterprise Threat Detection and Response

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AI in Cyber Defence – Blue Team

Objective and Project Description

To design and implement a mini Security Operations Center (SOC) that integrates AI-powered threat detection, SIEM, automated response, and defensive control mapping — simulating how a real enterprise defence team would monitor and respond to threats.

What was built

Windows Server 2022 was used as the monitored endpoint inside the Simplilearn Cybersecurity Lab. Security event logs were exported using a PowerShell script (generated with the help of ChatGPT) and ingested into Splunk Enterprise 9.4.0 as a searchable index.

SPL queries were then used to detect logon activity, privilege escalation and failed authentication attempts. A suspicious host IP was pivoted on and blocked automatically through PowerShell firewall rules — an automated, SOAR-style response. Finally, the defensive controls in place were mapped against the MITRE D3FEND framework to calculate a coverage score and identify the remaining security gap.

Tools and Industry Relevance

Splunk Enterprise 9.4.0	SIEM platform used for centralised log ingestion, indexing and SPL-based correlation and investigation.
ChatGPT (AI assistant)	Used to generate the PowerShell export script, the firewall automation script and to interpret detection results.
Windows PowerShell	Log extraction from the Security event channel, and automated firewall rule creation as the response action.
MITRE D3FEND	Defensive countermeasure framework used to map implemented controls and perform gap analysis.

Tasks Performed

1

Collect and ingest logs

Export Windows Security events to CSV and ingest into Splunk.

2

Augment with AI-driven detection

Run SPL queries for logon, privilege and failed-authentication events.

3

Threat hunting and IOC pivoting

Drill into privilege-escalation events and identify the responsible account.

4

Automated response and gap analysis

Block the suspicious IP via firewall automation and map controls to MITRE D3FEND.

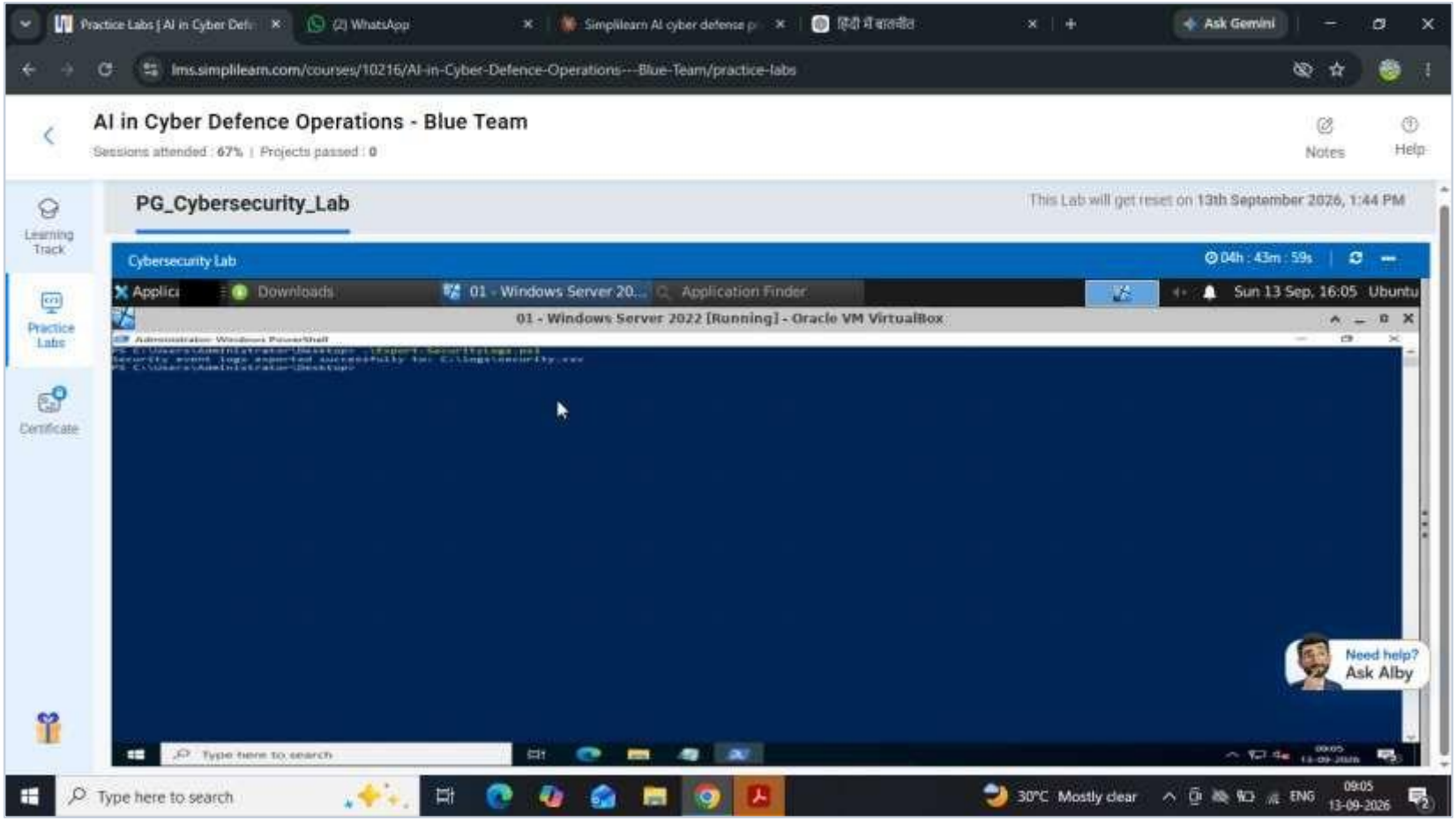
TASK 1

Collect and Ingest Logs

Windows Security event logs from the last seven days were exported to CSV using a PowerShell script and ingested into Splunk Enterprise as a searchable index, forming the data foundation for the entire SOC workflow.

PowerShell export of Security event logs

Task 1 — Log collection



Action performed

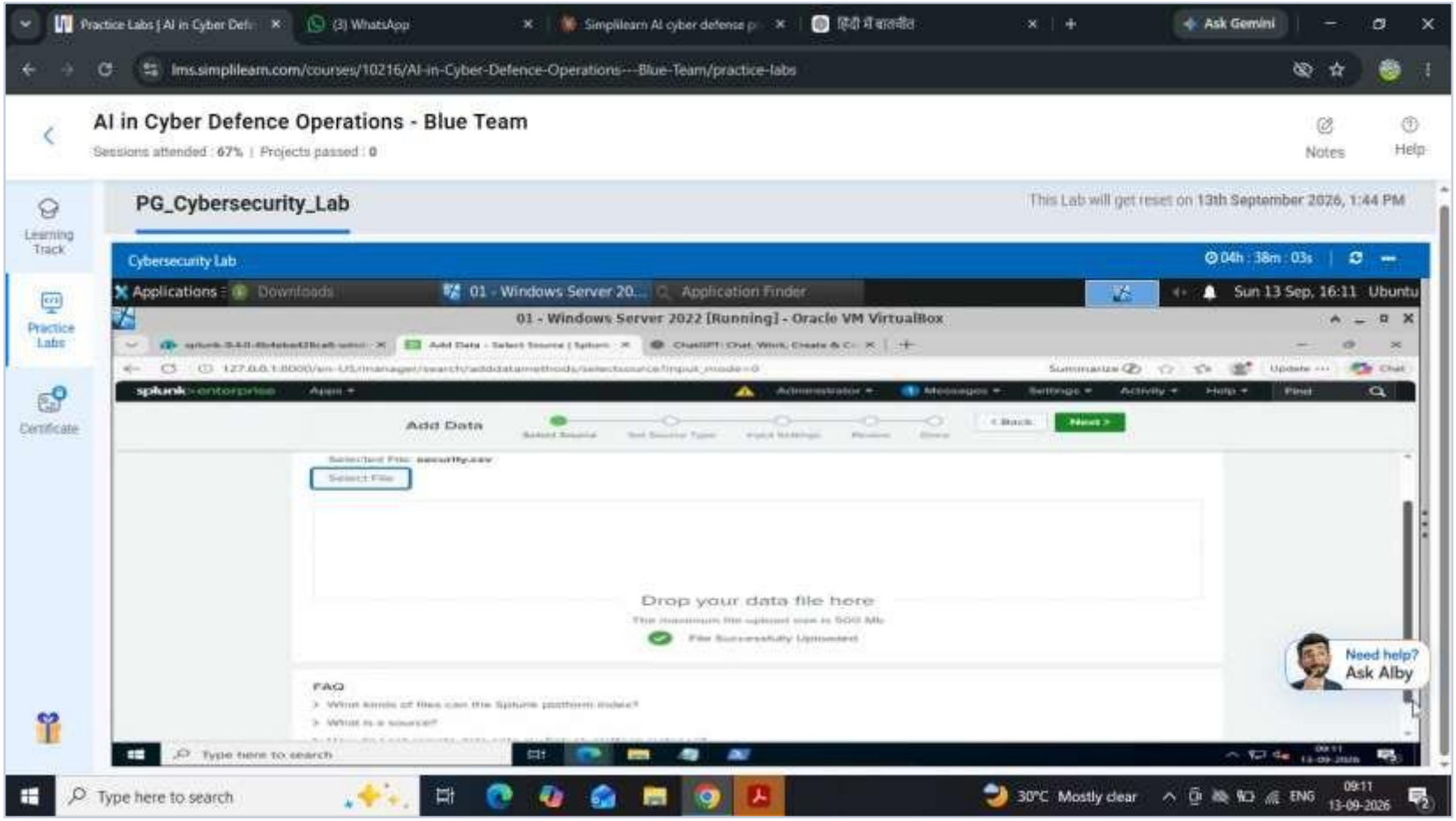
A PowerShell script (Export-SecurityLogs.ps1) was written with AI assistance and executed from an elevated PowerShell session on the Windows Server 2022 host. The script filters the Security event channel for the last seven days and writes the selected fields to CSV.

Observation

The script completed without error and returned the confirmation message “Security event logs exported successfully to: C:\logs\security.csv”. This confirms the export path and that the log collection stage worked as intended.

File selected for ingestion in Splunk

Task 1 — Log collection



Action performed

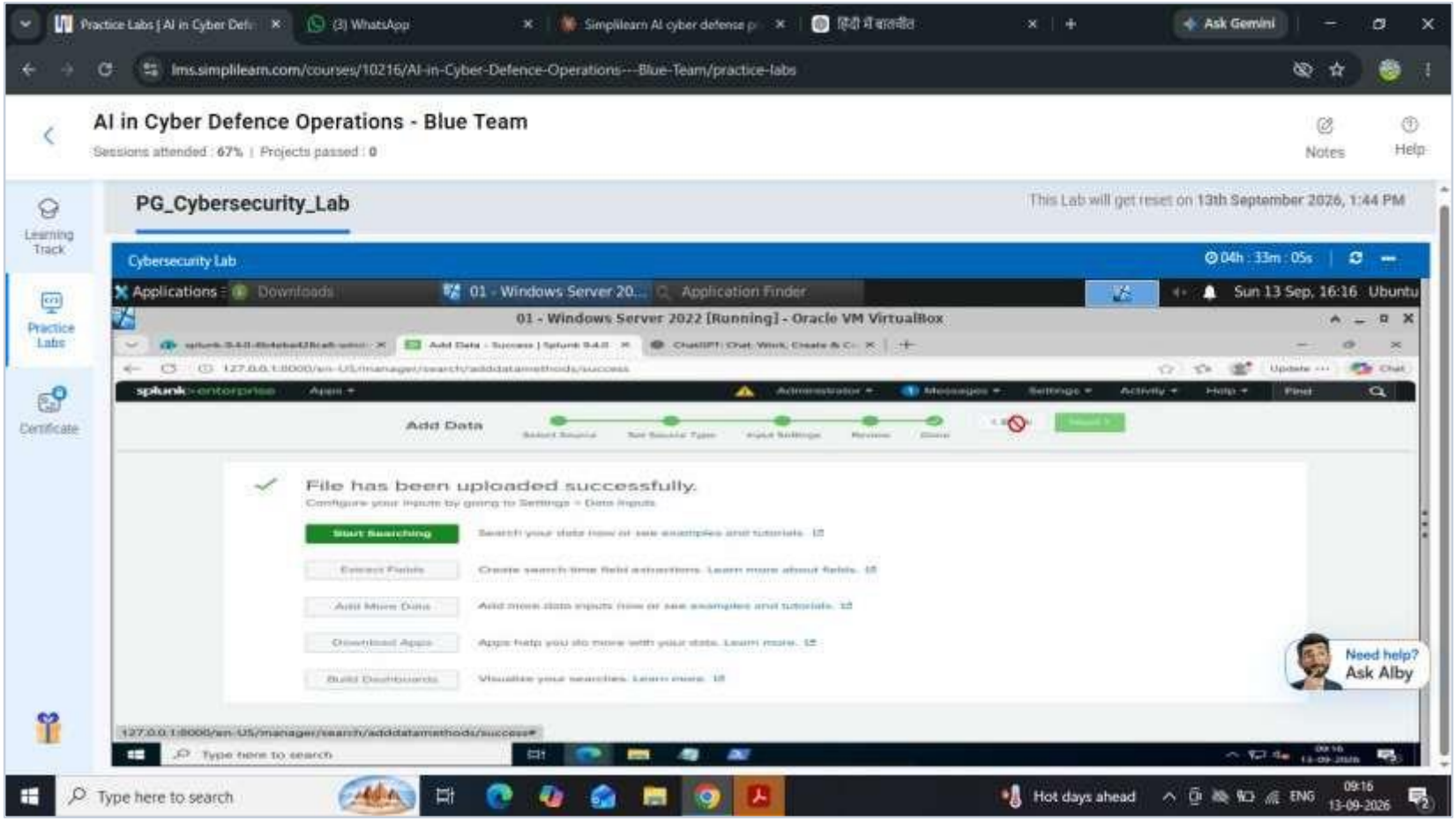
Inside Splunk, Settings → Add Data → Upload was used to select the exported security.csv file from C:\logs as the data source for indexing.

Observation

Splunk accepted the file and displayed “File Successfully Uploaded” with security.csv listed as the selected file, confirming the CSV was readable and within the 500 MB upload limit.

Data ingestion completed successfully

Task 1 — Log collection



Action performed

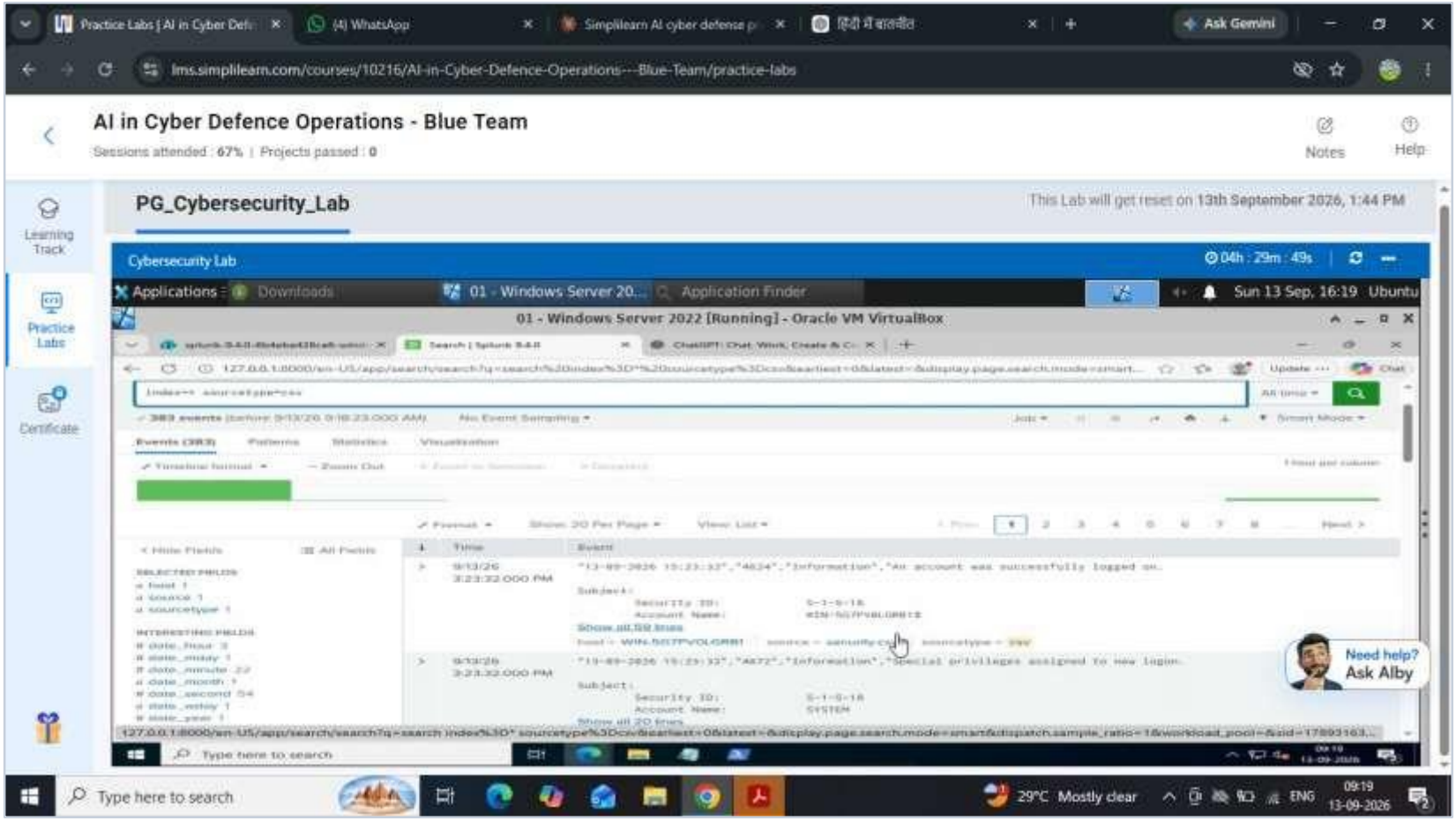
The Add Data wizard was completed through the Set Source Type, Input Settings and Review stages, with the data written into a dedicated index named security_logs.

Observation

The final confirmation screen shows “File has been uploaded successfully” with all five stages of the wizard marked complete, meaning the Windows logs are now indexed and searchable inside the SIEM.

Indexed events verified in Splunk search

Task 1 — Verification



Action performed

A baseline search (index=* sourcetype=csv) was run over an All-time window to verify that the ingested records were searchable and correctly parsed.

Observation

Splunk returned 383 events. The raw event text is correctly parsed — Event ID 4624 (“An account was successfully logged on”) is visible along with Security ID, Account Name and the host WIN-5G7PV0LGRB1, confirming field-level readability.

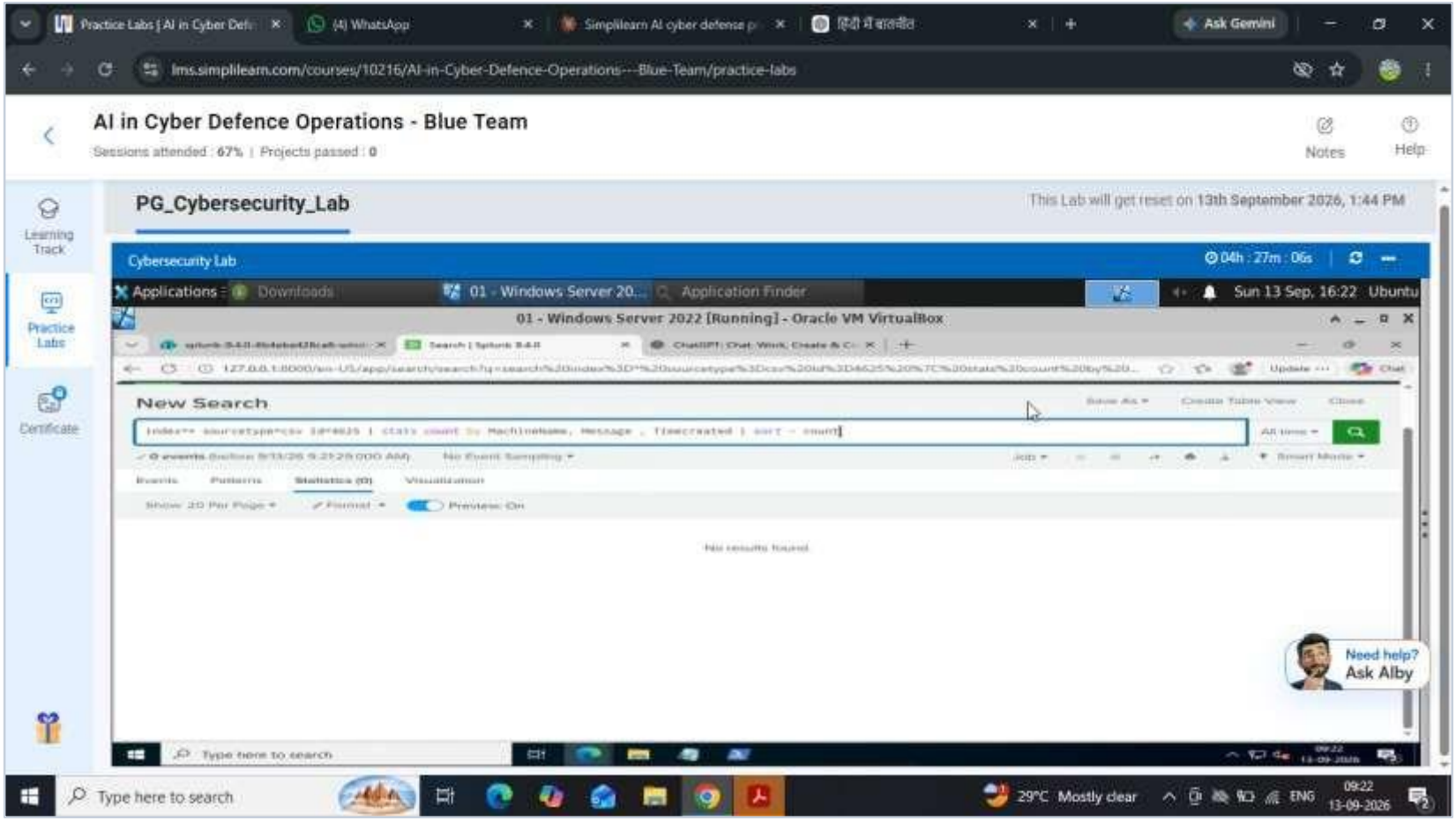
TASK 2

AI-Driven Detection and Threat Intelligence

SPL detection queries generated with AI assistance were run against the indexed data to identify authentication activity, privilege assignment and any failed logon attempts, and to build an overall picture of event distribution across the host.

Failed logon detection (Event ID 4625)

Task 2 — Detection



Action performed

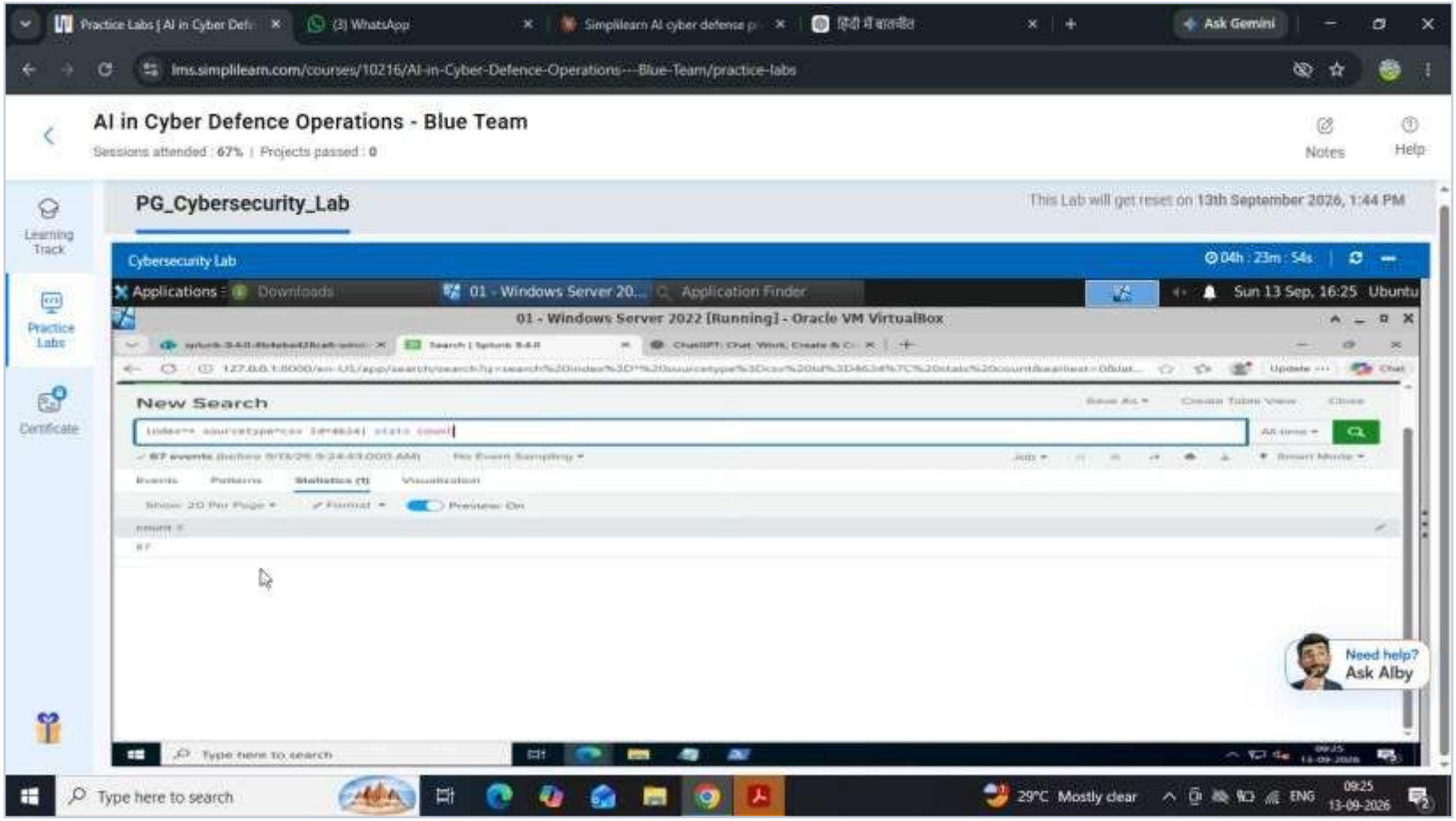
An SPL query was run to isolate Event ID 4625, which Windows records whenever an account fails to log on. This is the primary indicator used to detect brute-force and password-spray attempts.

Observation

The query returned zero events. No failed logon attempts occurred on this host during the seven-day window, so no brute-force or credential-guessing activity was present. A nil result is still a valid detection outcome and is reported as such.

Successful logon events (Event ID 4624)

Task 2 — Detection



Action performed

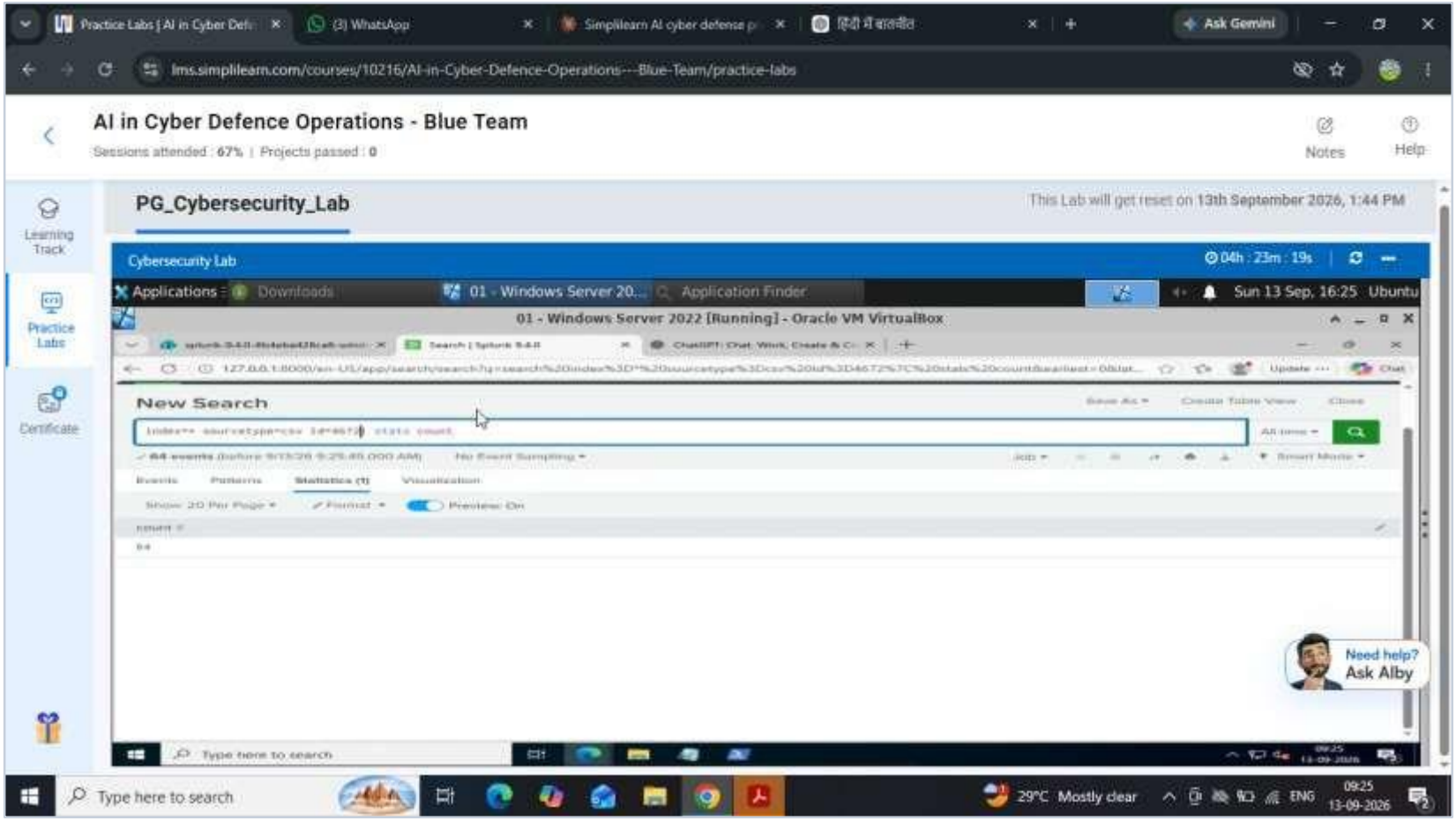
Event ID 4624 was counted to establish the volume of successful authentications against the monitored server during the collection period.

Observation

67 successful logon events were recorded. With zero failures alongside 67 successes, the authentication pattern is consistent with normal service and system activity rather than an attacker attempting access.

Privilege assignment events (Event ID 4672)

Task 2 — Detection



Action performed

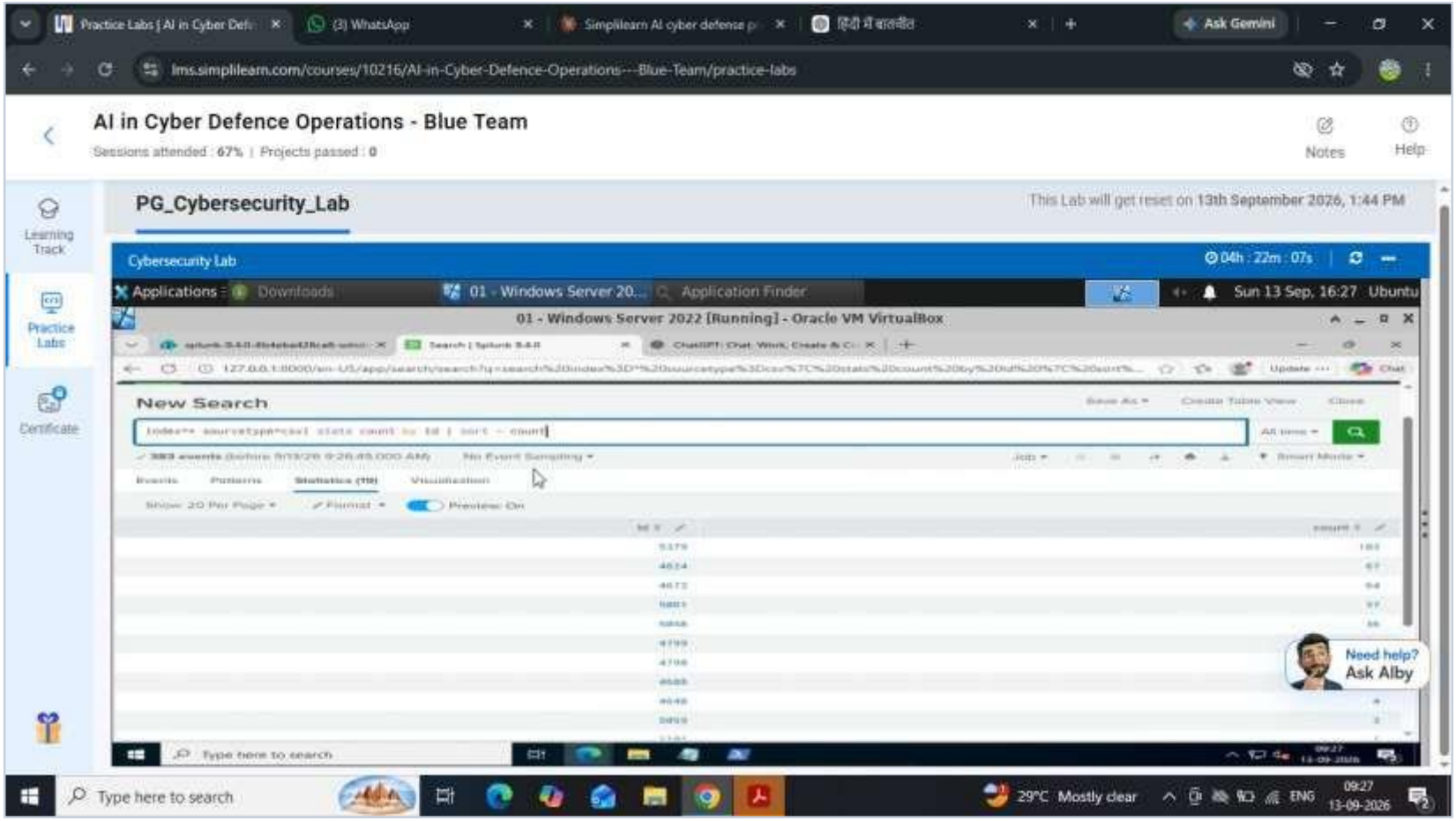
Event ID 4672 — “Special privileges assigned to new logon” — was counted. This event is a high-value detection signal because it fires whenever an account is granted administrative-level privileges.

Observation

64 privilege-assignment events were found. A count this close to the total logon count warrants investigation, so this became the lead indicator carried forward into threat hunting in Task 3.

Event distribution across the host

Task 2 — Detection



Action performed

A summary query (stats count by Id | sort - count) was run to profile every distinct Event ID present in the dataset and rank them by frequency.

Observation

19 distinct Event IDs were identified across 383 events. The top entries are 5379 (credential manager read, 103), 4624 (successful logon, 67) and 4672 (special privileges, 64), giving a clear baseline of what normal activity looks like on this host.

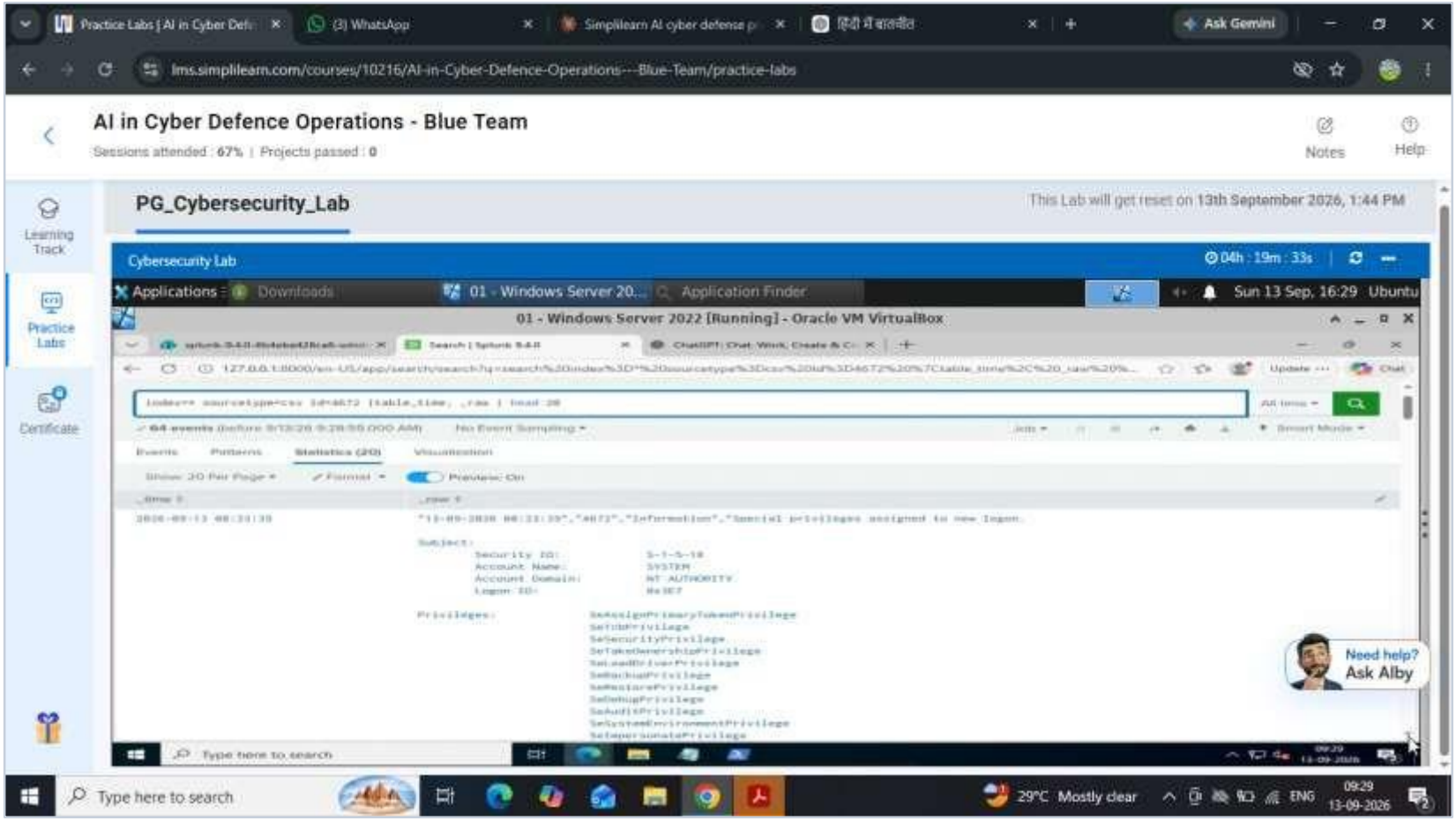
TASK 3

Threat Hunting and IOC Pivoting

The privilege-escalation signal identified in Task 2 was pivoted on and examined at raw-event level to determine which account was responsible and whether the behaviour represented a genuine compromise or legitimate system activity.

Pivot into privilege escalation events

Task 3 — Threat hunting



Action performed

The 64 Event ID 4672 records were pivoted into a raw-event table to inspect the Subject block — Security ID, Account Name, Account Domain and the exact privileges granted on each occurrence.

Observation

Every event resolves to Security ID S-1-5-18, Account Name SYSTEM, domain NT AUTHORITY — the built-in local System account. The privileges listed (SeTcbPrivilege, SeDebugPrivilege, SeBackupPrivilege and others) are those Windows assigns to its own services at boot. The activity is therefore legitimate, not an intrusion. No external malicious IOC was present in this dataset, so no third-party threat-intelligence enrichment was applicable.

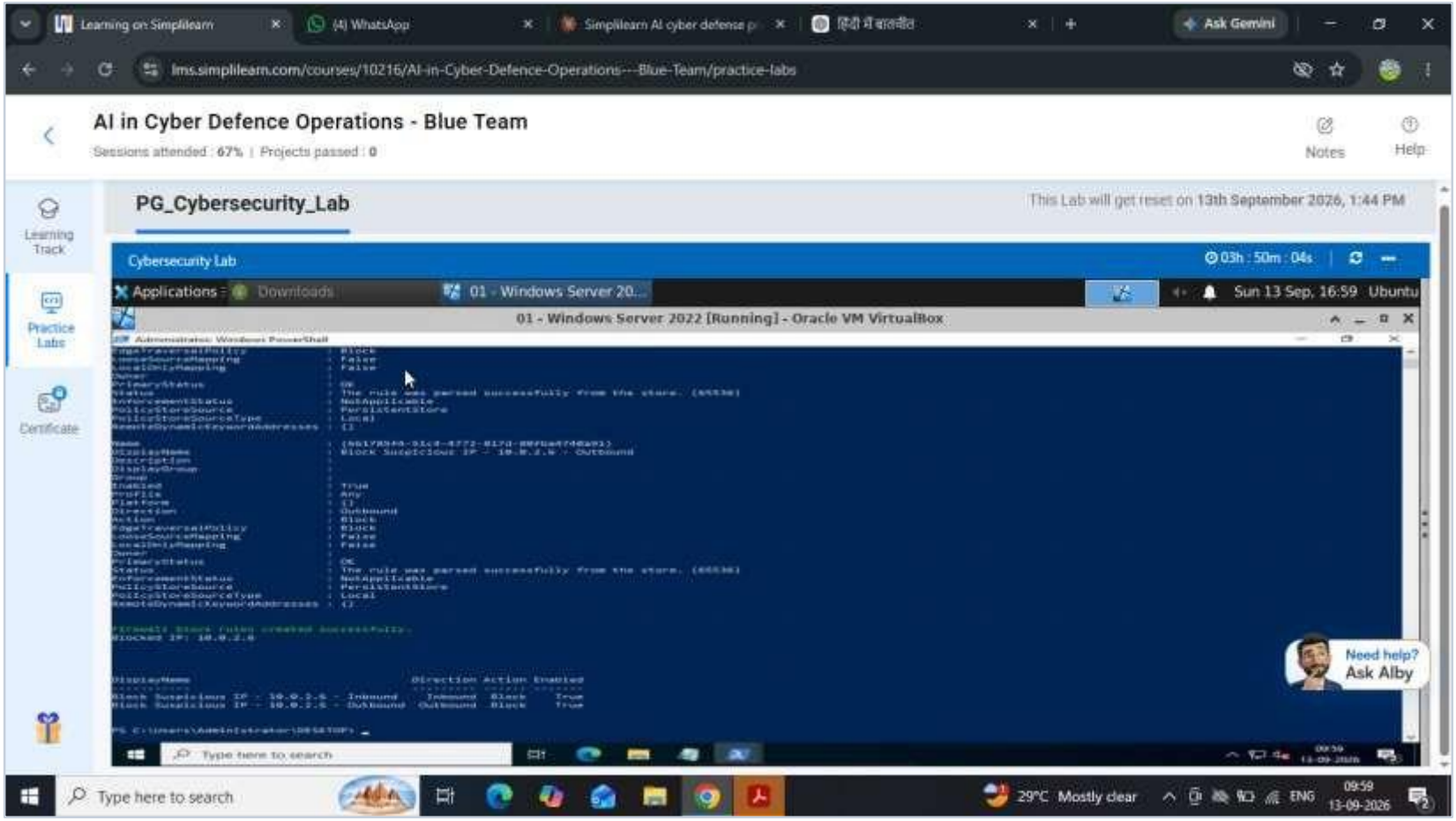
TASK 4

Automated Response and MITRE D3FEND Gap Analysis

A SOAR-style automated containment action was executed against the identified host IP, and the defensive controls implemented across the environment were mapped to the MITRE D3FEND framework to produce a measurable coverage score.

Automated firewall containment executed

Task 4 — Automated response



Action performed

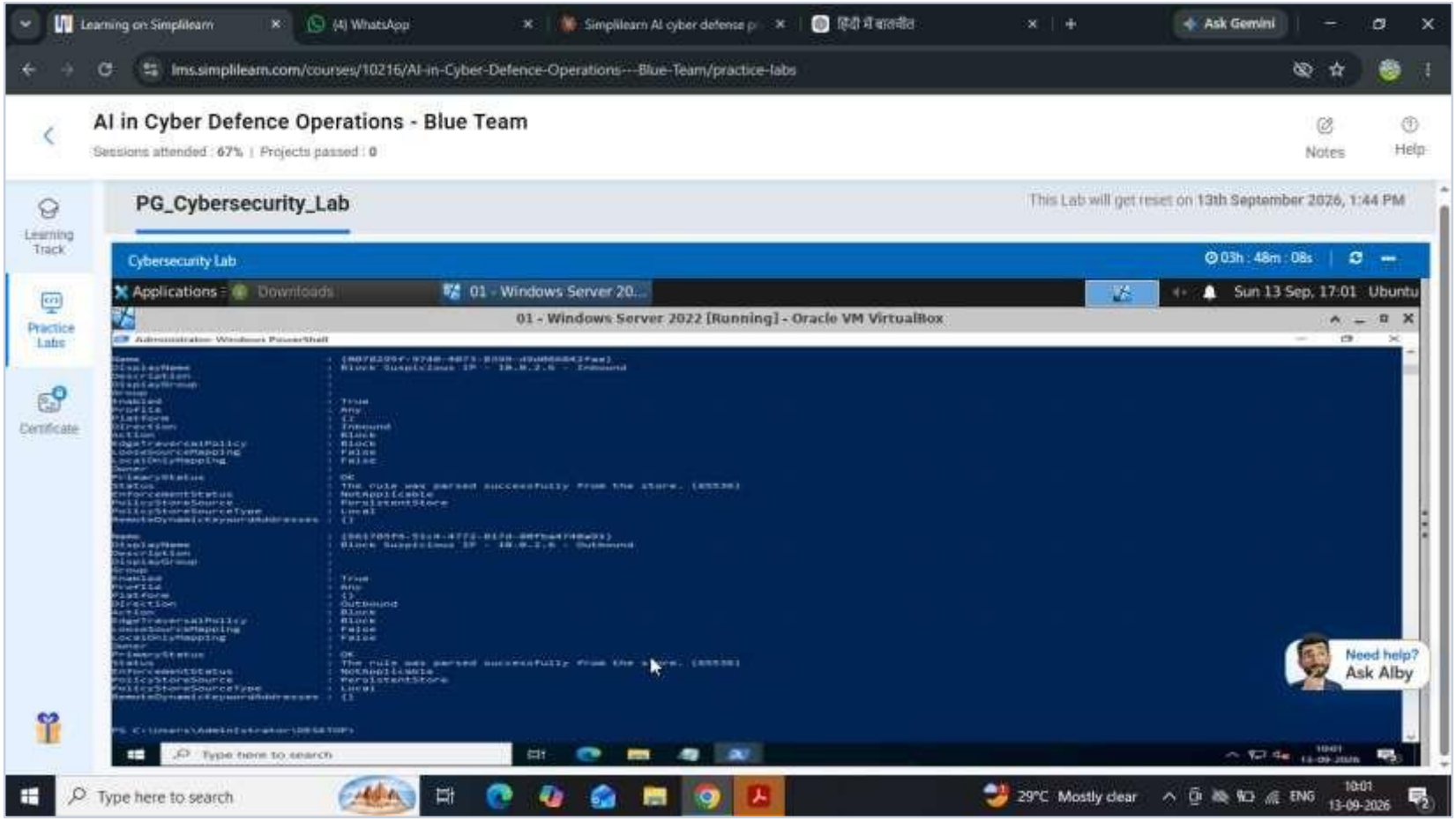
A PowerShell script generated with AI assistance was run to create Windows Defender Firewall rules blocking all traffic to and from the suspicious host address 10.0.2.6 — replicating the containment action a SOAR platform would trigger automatically on alert.

Observation

The script reported “Firewall block rules created successfully. Blocked IP: 10.0.2.6” and both rules — inbound and outbound — are listed with Action = Block and Enabled = True. Containment was achieved in seconds without manual console work.

Verification of the created firewall rules

Task 4 — Automated response



Action performed

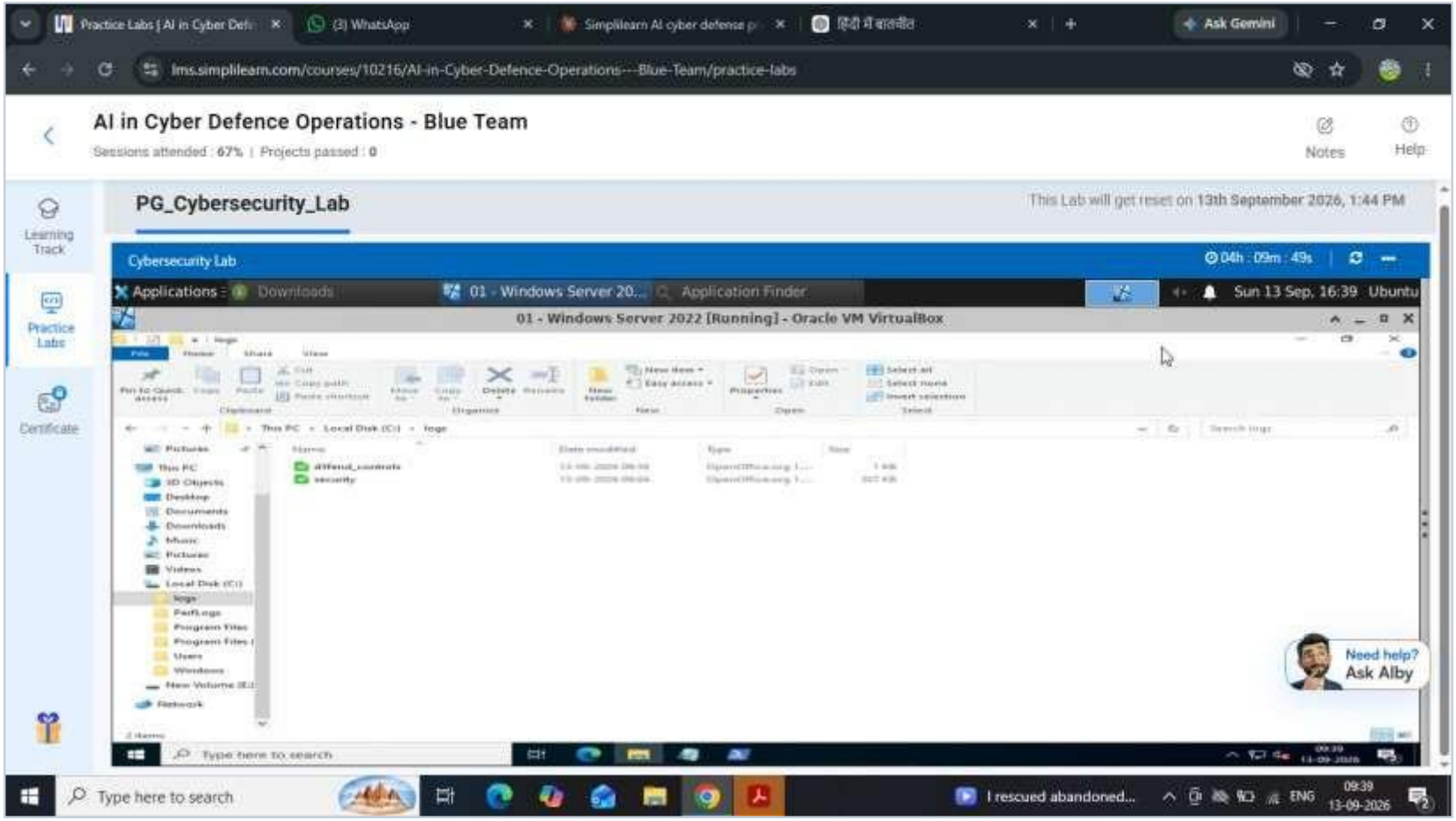
Get-NetFirewallRule was used to independently verify that both rules persisted correctly in the firewall policy store rather than relying on the script's own output.

Observation

Both rules return PrimaryStatus OK with the status “The rule was parsed successfully from the store”, PolicyStoreSource PersistentStore and Enabled True. This confirms the block survives a reboot and is genuinely enforced by the host firewall.

D3FEND control inventory prepared

Task 4 — Control mapping



Action performed

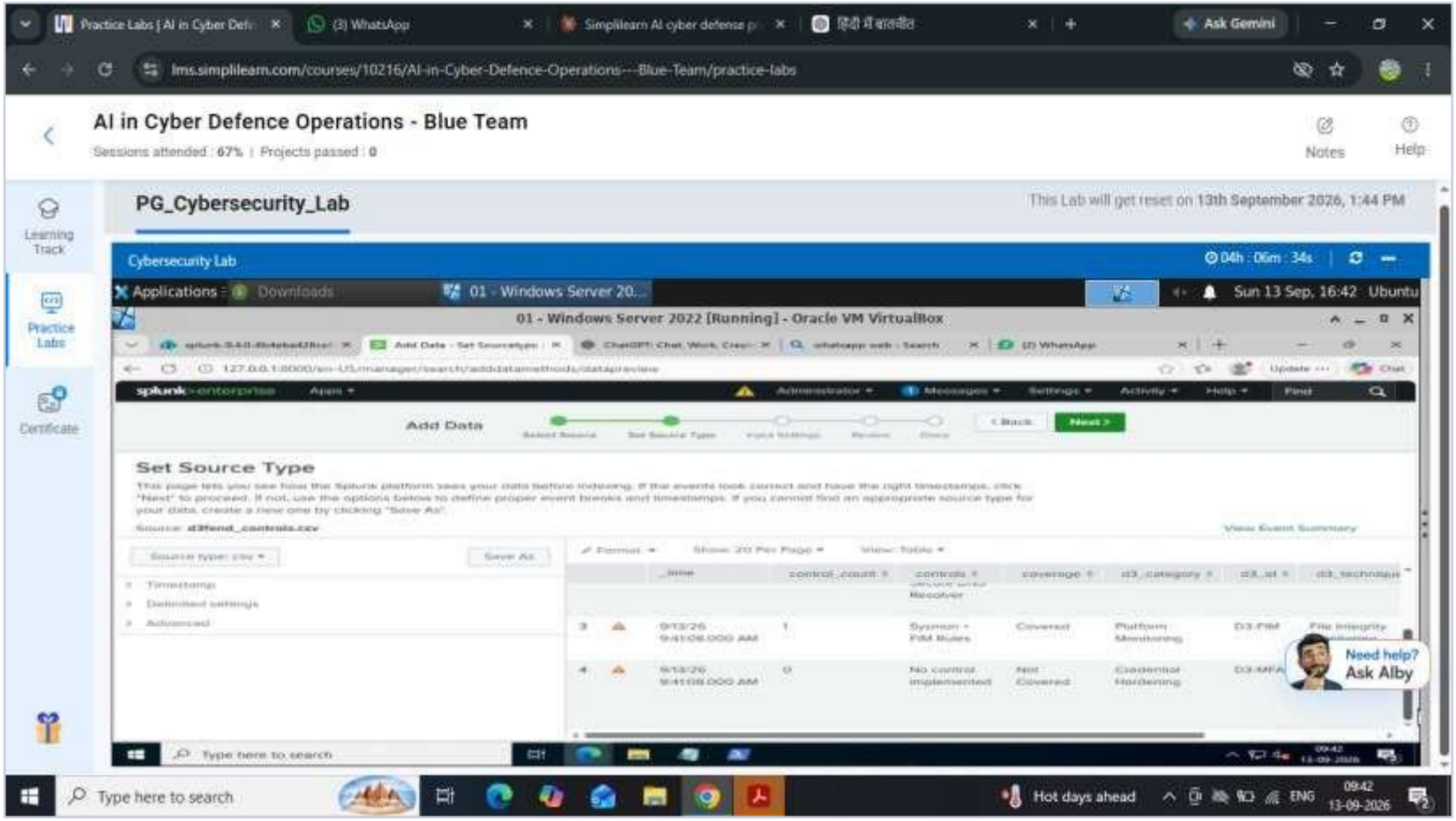
A control inventory (d3fend_controls.csv) was authored, listing each implemented defensive measure against its corresponding MITRE D3FEND technique ID, category and coverage status. It was saved alongside the log export in C:\logs.

Observation

Both datasets are now in place — security.csv (307 KB of event data) and d3fend_controls.csv (the control mapping) — ready to be correlated inside the same SIEM instance.

Control mapping parsed by Splunk

Task 4 — Control mapping



Action performed

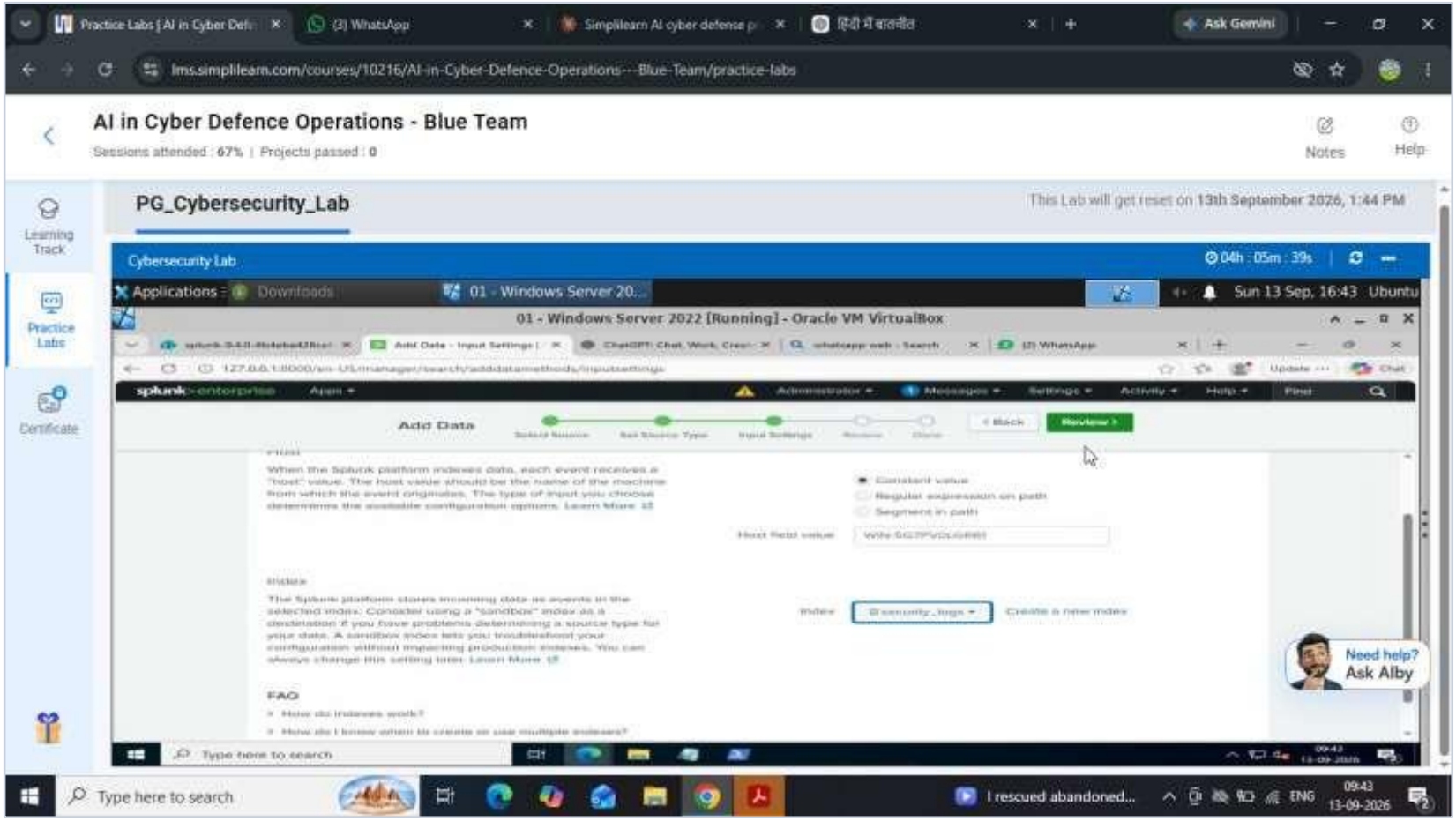
The control inventory was uploaded through Add Data and the Set Source Type preview was used to confirm Splunk had parsed the file into distinct fields before committing it to an index.

Observation

All expected columns resolved correctly — control_count, controls, coverage, d3_category, d3_id and d3_technique. Row 4 already shows the gap: Credential Hardening / D3-MFA with “No control implemented” and coverage “Not Covered”.

Index selection for the control dataset

Task 4 — Control mapping



Action performed

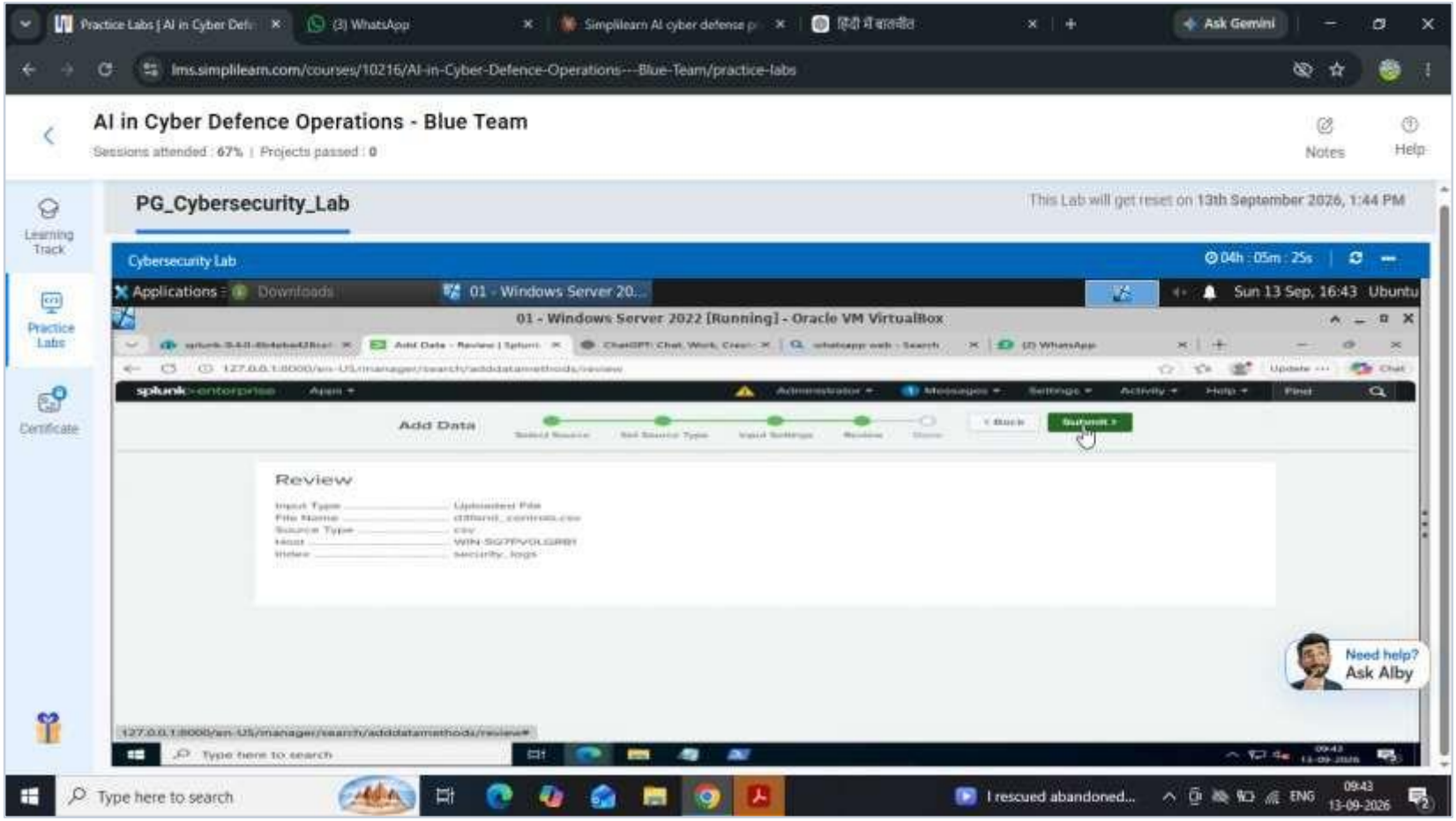
In Input Settings the control mapping was directed into the existing security_logs index, so the defensive-control data sits alongside the event data and can be queried in the same search context.

Observation

Host field value resolves to WIN-5G7PV0LGRB1 and the index is set to security_logs, keeping both datasets under a single searchable index rather than fragmenting them.

Ingestion settings reviewed before submission

Task 4 — Control mapping



Action performed

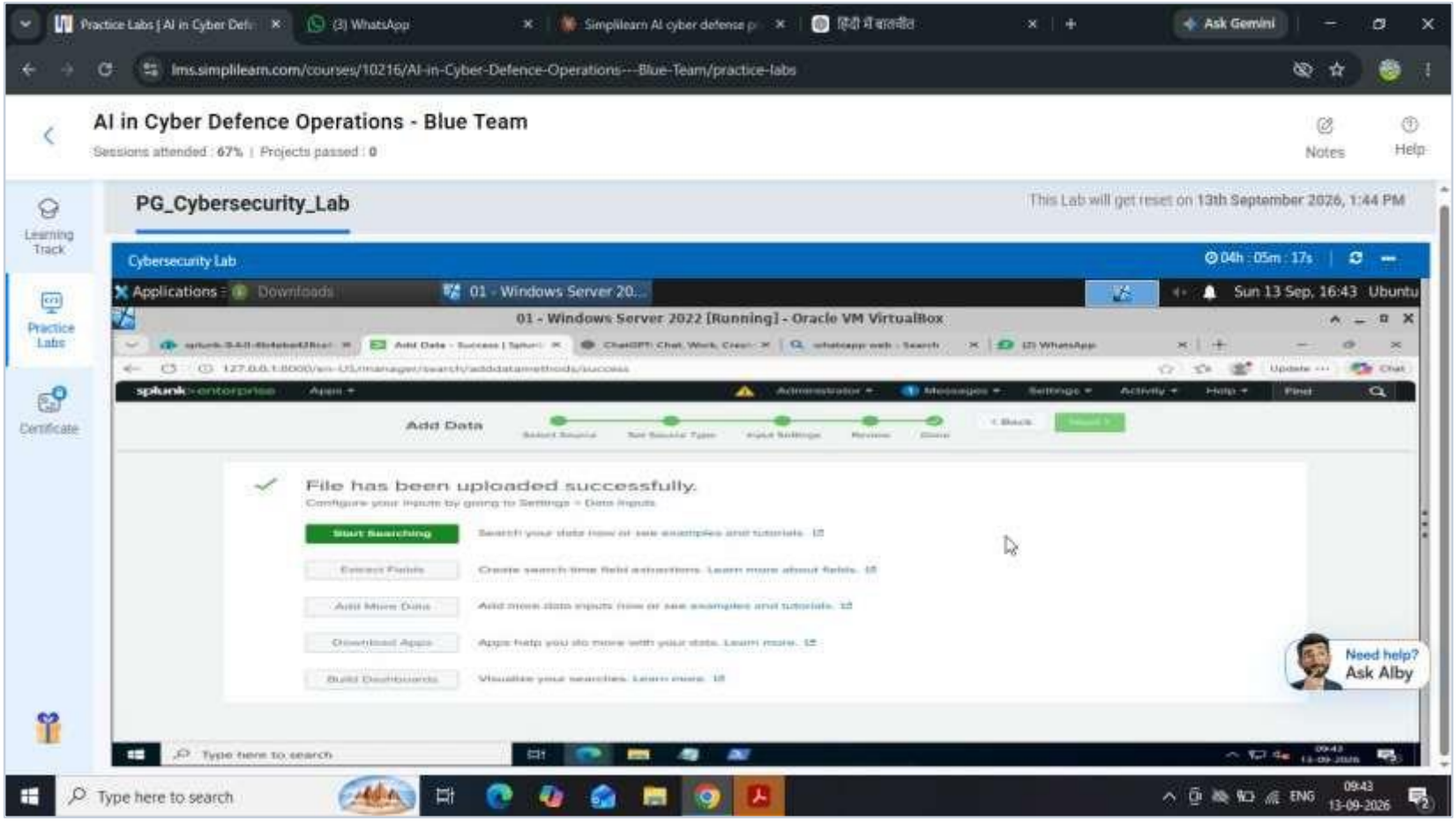
The Review stage was used to confirm the input configuration — input type, file name, source type, host and destination index — before the data was committed.

Observation

All values are correct: uploaded file d3fend_controls.csv, source type csv, host WIN-5G7PV0LGRB1, index security_logs. Reviewing before submission avoids having to clean up a mis-indexed dataset afterwards.

Control dataset ingestion completed

Task 4 — Control mapping



Action performed

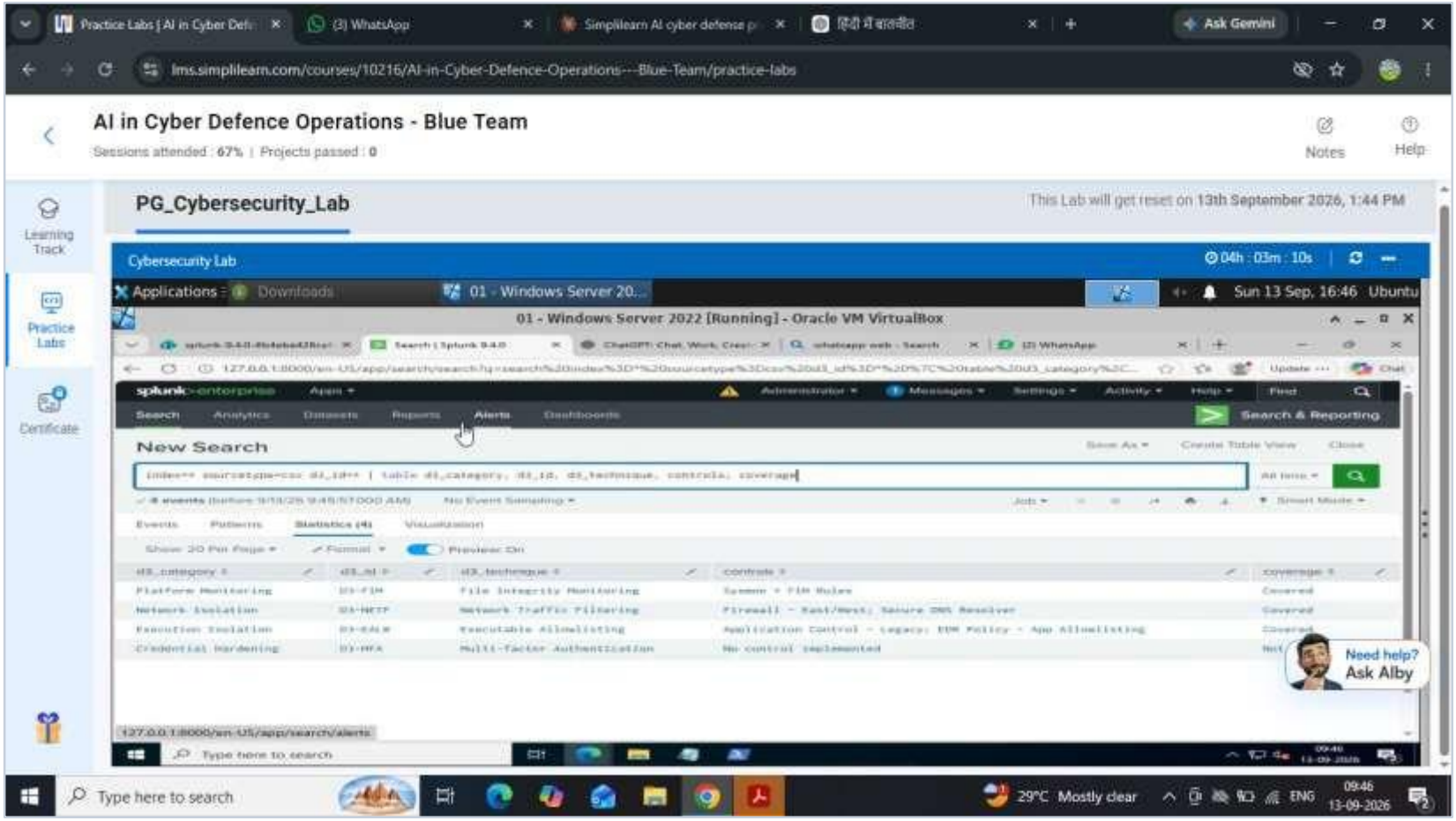
The ingestion was submitted and Splunk completed the final Done stage of the Add Data wizard.

Observation

The confirmation “File has been uploaded successfully” is returned with every stage marked complete, meaning the D3FEND control mapping is now queryable through SPL.

MITRE D3FEND coverage mapping

Task 4 — Gap analysis



Action performed

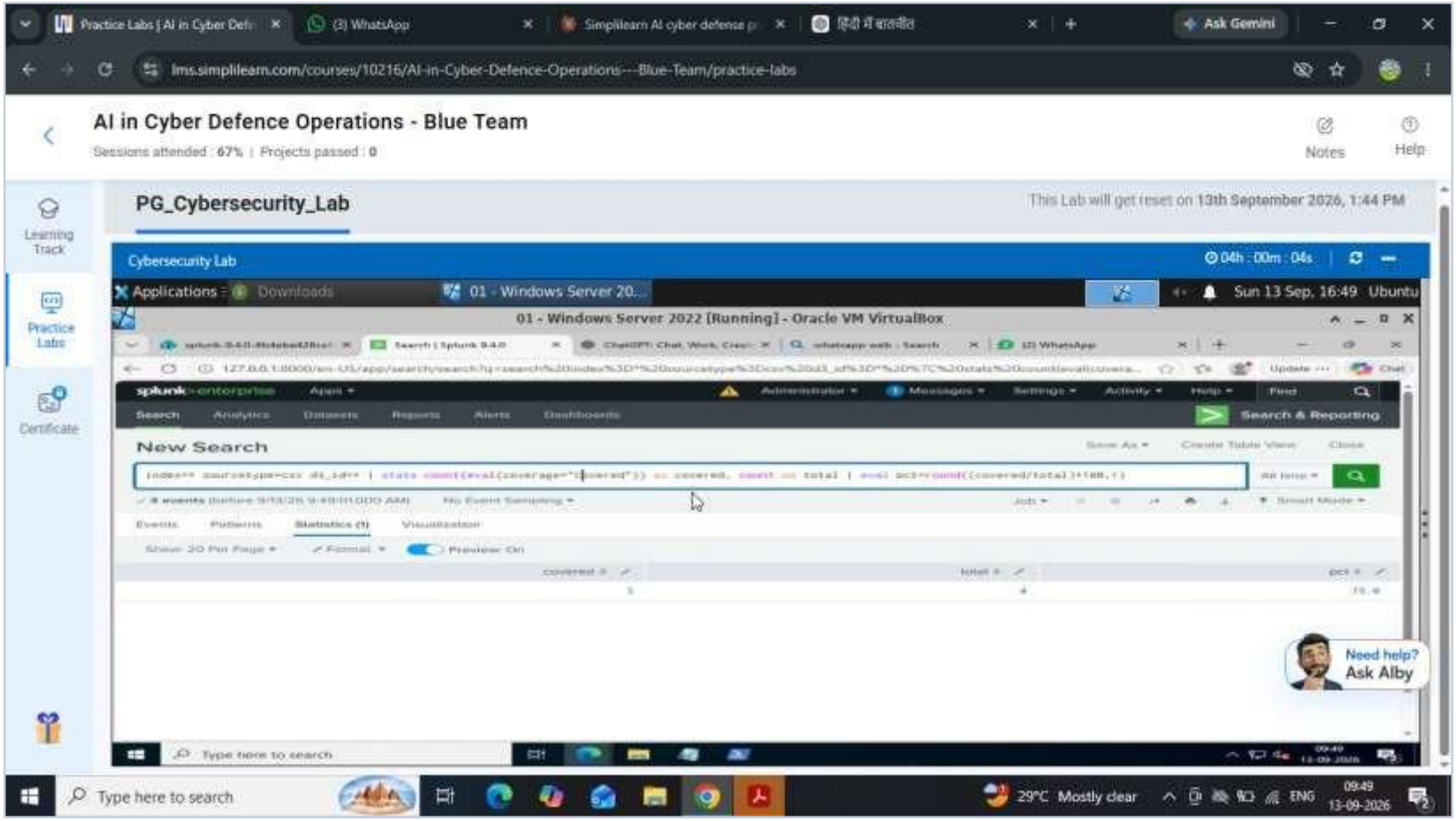
An SPL query was run to render the full control mapping as a table — each D3FEND category alongside its technique ID, the specific controls implemented and the resulting coverage status.

Observation

Four D3FEND techniques are mapped. D3-FIM (File Integrity Monitoring), D3-NETF (Network Traffic Filtering) and D3-EALW (Executable Allowlisting) are Covered by Sysmon, firewall and application-control policies respectively. D3-MFA (Multi-factor Authentication) is Not Covered — the single identified gap.

Defensive coverage score calculated

Task 4 — Gap analysis



Action performed

A statistical query was used to compute the coverage score automatically — counting techniques marked Covered against the total mapped, then deriving the percentage.

Observation

The result is 3 covered out of 4 total, giving a defensive coverage score of 75.0%. The remaining 25% corresponds to the missing multi-factor authentication control, which is the concrete, evidence-backed recommendation coming out of this assessment.

Key Findings

383	Total security events ingested and indexed from the monitored host
0	Failed logon attempts (Event ID 4625) — no brute-force activity detected
67 / 64	Successful logons and privilege-assignment events, all attributable to SYSTEM
2	Firewall rules created automatically to contain the suspicious host address
75%	MITRE D3FEND defensive coverage — 3 of 4 mapped techniques covered

Conclusion and Recommendations

Conclusion

A functional mini-SOC was built end to end: logs were collected from a live Windows Server, ingested into Splunk, analysed through AI-assisted SPL detection queries, hunted at raw-event level, contained through automated firewall response, and finally measured against the MITRE D3FEND framework. The environment showed no evidence of compromise — all privilege activity traced back to the legitimate SYSTEM account — but the exercise did surface a real, actionable control gap.

Recommendations

- Implement multi-factor authentication (D3-MFA) for privileged logons to close the 25% coverage gap.
- Move from manual CSV export to a Splunk Universal Forwarder for continuous, real-time log ingestion.
- Convert the detection queries used here into scheduled Splunk alerts so they fire without manual execution.
- Extend the D3FEND mapping across more categories to build a fuller picture of defensive posture.